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MID TERM

MTH101 - Calculus And Analytical Geometry

Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative ..... At x_0

- Minima
- Maxima
- None of these

Question No: 2 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative ..... At x_0

- Minima
- Maxima
- None of these

Question No: 3 (Marks: 1) - Please choose one

A line $y = y_0$ is called a ..... for the graph f if

$$\lim_{x \to +\infty} f(x) = y_0 \quad or \quad \lim_{x \to -\infty} f(x) = y_0$$

- Vertical asymptotes
- Horizontal asymptotes
- None of these

Question No: 4 (Marks: 1) - Please choose one

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

- $3x^7 + 2$
- $24x^7 + 2$
- $3x^9 + 2x^2$
- $24x^9 + 2x^2$

Question No: 5 (Marks: 1) - Please choose one

$\frac{d(\sec x)}{dx} =$

- $(\sec x)(\tan x)$
- $(\sec x)(\tan x)$
- $(co\sec x)(co\,t\,x)$

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$(co\sec x)(\tan x)$

Question No: 6 (Marks: 1) - Please choose one

If $y=\frac{1}{1-x}$ then $\frac{dy}{dx}=$

- 1
- -1
- $\frac{1}{(1-x)^2}$
- $\frac{-1}{(1-x)^2}$

Question No: 7 (Marks: 1) - Please choose one

If $xy=4$ then $\frac{dy}{dx}=$

- 0
- $\frac{-1}{x^2}$
- $\frac{4}{x^2}$
- $\frac{-4}{x^2}$

Question No: 8 (Marks: 1) - Please choose one

$\frac{d}{dx}[\text{cosec } x]=$ _____

- $\frac{1}{1+\cos^2 x}$
- $\frac{-\cos x}{1-\cos^2 x}$
- $\frac{-\cos x}{1-\cos^2 x}$
- $\frac{1}{1-\cos^2 x}$

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Question No: 9 (Marks: 1) - Please choose one

$y = f\left(g\left(h(x)\right)\right)$

If

$u = g\left(h(x)\right)$

$v = h(x)$ $\frac{dy}{dx} =$ _____

then

- $\frac{dy}{du}.\frac{du}{dv}.\frac{dv}{dx}$
- $\frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}$
- $\frac{dv}{du}.\frac{du}{dv}.\frac{dy}{dx}$

Question No: 10 (Marks: 1) - Please choose one

$\frac{d}{dx}[f(g(x))] =$

- $[f'(g(x))]\ [f'(x)]$
- $f'(g(x)) + g'(x)$
- $[f'(g(x))]\ [g'(x)]$
- $f'(g(x)) - g'(x)$

Question No: 11 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- Product
- Sum
- Difference
- Composition

Question No: 12 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- f is an increasing function.
- f is a decreasing function.
- f is a constant function.

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Question No: 13 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- f is an increasing function.
- f is a decreasing function.
- f is a constant function.

Question No: 14 (Marks: 1) - Please choose one

If $f'(x) < 0$ on an open interval (a,b) then which of the following statement is correct?

- f is concave up on (a, b).
- f is concave down on (a, b)
- f is linear on (a, b).

Question No: 15 (Marks: 1) - Please choose one

$If\ x > 0\ then\ \frac{d}{dx}[\ln x] = _____$

- 1
- x
- $\frac{1}{x}$
- $\ln \frac{1}{x}$

Question No: 16 (Marks: 1) - Please choose one

$If\ b > 0\ then\ \frac{d}{dx}[b^x] = _____$

- 0
- xb^{x-1}
- $\ln b$
- $b^x \ln b$

Question No: 17 (Marks: 1) - Please choose one

$If\ \ x > \frac{1}{2}\ \ then\ \ \frac{d}{dx}[\ln 2x] = _____$

- $\frac{1}{x}$

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- $\frac{2}{x}$
- $\frac{1}{2x}$
- $\frac{2}{x^2}$

Question No: 18 (Marks: 1) - Please choose one

Let $y=(x^3+2x)^{37}$. Which of the following is correct?

- $\frac{dy}{dx}=(37)(x^3+2x)^{36}$
- $\frac{dy}{dx}=111x^2(x^3+2x)^{36}$
- $\frac{dy}{dx}=(111x^2+74)(x^3+2x)^{36}$
- $\frac{dy}{dx}=(111x^2+74)(x^3+2x)^{38}$

Question No: 19 (Marks: 1) - Please choose one

What is the base of natural logarithm?

- 2.71
- 10
- 5
- Any real number

Question No: 20 (Marks: 1) - Please choose one

Let x_0 be critical points of the function f. Those critical points for which $f'(x_0)=0$ are called _____ of f

- Local points
- End points
- Stationary points

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Question No: 21 (Marks: 1) - Please choose one

If we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} =$ _____

- $\frac{-x}{y}$
- $\frac{x}{y}$
- $\frac{-y}{x}$
- None of these

Question No: 22 (Marks: 1) - Please choose one

$\log_b ac =$ _____

- $\log_b a + \log_b c$
- $\log_b a - \log_b c$
- $\frac{\log_b a}{\log_b c}$
- $(\log_b a)(\log_b c)$

Question No: 23 (Marks: 1) - Please choose one

$\log_b a^r =$ _____

- $a \log_b r$
- $r \log_b a$
- $\frac{\log_b a}{\log_b r}$
- $\log_b a + \log_b r$

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Question No: 24 (Marks: 1) - Please choose one

$\log_b \frac{1}{t} =$ _____

- $\log_b t$
- $1-\log_b t$
- $1+\log_b t$
- $-\log_b t$

Question No: 25 (Marks: 3)

Find $\frac{dy}{dx}$ if $y = x^{\frac{p}{q}}$

Question No: 26 (Marks: 5)

Differentiate $\sqrt{13x^2-5x+8}$

Question No: 27 (Marks: 10)

Determine the intervals in which the graph of the function $f(x)=\frac{6}{x^2+3}$ is concave upward or downward

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